

LAPORAN WORKSHOP SISTEM OPERASI
PERTEMUAN 13 - KONFIGURASI JARINGAN DI UBUNTU

Politeknik Caltex Riau

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PROGRAM STUDI D4 – SISTEM INFORMASI
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK CALTEX RIAU

TUGAS PRAKTIKUM

Konfigurasi Jaringan Di Ubuntu

Percobaan Praktikum

Percobaan 1 : Mengecek Informasi Network

\$ ip a : me-gecek “Biodata + identitas” dalam jaringan. address=alamat yang dipakai komputer untuk dikenali dalam jaringan. [Ipnya 127.0.0.1/ 8]

NOTE: Ip di lab sama karna : sharing alamat

```
shoflyyah@SHOFI2SIEUbuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b6:61:72 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83759sec preferred_lft 83759sec
    inet6 fd17:625c:f037:2:dba5:1e0c:ec8:90b8/64 scope global temporary dynamic
        valid_lft 86132sec preferred_lft 14132sec
    inet6 fd17:625c:f037:2:6cfa:f70e:d718:140f/64 scope global dynamic mngtmpdr noprefixroute
        valid_lft 86132sec preferred_lft 14132sec
    inet6 fe80::e854:813:4dae:5f98/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

\$ ip route : untuk melihat rute/jalan yang digunakan komputer agar bisa mengirim data ke tujuan.

```
shoflyyah@SHOFI2SIEUbuntu:~$ ip r
default via 10.0.2.2 dev enp0s3 proto dhcp metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
169.254.0.0/16 dev enp0s3 scope link metric 1000
shoflyyah@SHOFI2SIEUbuntu:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
169.254.0.0/16 dev enp0s3 scope link metric 1000
```

\$ systemd-resolve --status : melihat konfigurasi DNS yang digunakan

```
shoflyyah@SHOFI2SIEUbuntu:~$ systemd-resolve --status
Global
    LLMNR setting: no
    MulticastDNS setting: no
    DNSOverTLS setting: no
    DNSSEC setting: no
    DNSSEC supported: no
    DNSSEC NTA: 10.in-addr.arpa
                16.172.in-addr.arpa
                168.192.in-addr.arpa
                17.172.in-addr.arpa
                18.172.in-addr.arpa
                19.172.in-addr.arpa
                20.172.in-addr.arpa
                21.172.in-addr.arpa
                22.172.in-addr.arpa
```

Percobaan 2 : Konfigurasi IP DHCP (Otomatis)

\$ su - : masuk sebagai user root (administrator penuh di Linux)

\$ sudo nano /etc/netplan/01-network-manager-all.yaml : Untuk mengedit pengaturan jaringan seperti IP, gateway, dan DNS menggunakan editor nano.

```
shofiyyah@SHOFI2SIEUbuntu:~$ su -
Password:
root@SHOFI2SIEUbuntu:~# sudo nano /etc/netplan/01-network-manager-all.yaml

GNU nano 4.8 /etc/netplan/01-network-manager-all.yaml Modified
# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: true

File Name to Write: /etc/netplan/01-network-manager-all.yaml
^G Get Help      M-D DOS Format  M-A Append     M-B Backup File
^C Cancel        M-M Mac Format  M-P Prepend    ^T To Files
```

jika sudah, ctrl +S dan ctrl + X

\$ sudo netplan apply : Menerapkan perubahan konfigurasi jaringan yang telah dibuat.

```
root@SHOFI2SIEUbuntu:~# sudo netplan apply
root@SHOFI2SIEUbuntu:~#
```

\$ ip a : Mengecek hasil konfigurasi jaringan (IP, interface, status).

```

root@SHOFI2SIEUbuntu:~# sudo netplan apply
root@SHOFI2SIEUbuntu:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:b6:61:72 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86128sec preferred_lft 86128sec
    inet6 fe80::a00:27ff:feb6:6172/64 scope link
        valid_lft forever preferred_lft forever
root@SHOFI2SIEUbuntu:~#
root@SHOFI2SIEUbuntu:~# ip r
default via 10.0.2.2 dev enp0s3 proto dhcp metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
169.254.0.0/16 dev enp0s3 scope link metric 1000
root@SHOFI2SIEUbuntu:~#

```

Percobaan 3 : Konfigurasi IP Static (Manual)

\$ su - : masuk sebagai user root (administrator penuh di Linux)

\$ sudo nano /etc/netplan/01-network-manager-all.yaml : membuka file konfigurasi jaringan utlk di edit.

```

GNU nano 4.8 /etc/netplan/01-network-manager-all.yaml
# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 10.0.2.15/24
      gateway4: 10.0.2.2
      nameservers:
        addresses:
          - 8.8.8.8
          - 1.1.1.1

```

[Read 14 lines]

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell

Addresess diambil dari inet yg ada di ip a

Kalo gateway4 dari ip r yg via 10.0.2.2

\$ sudo netplan apply : Menerapkan perubahan konfigurasi jaringan yang telah dibuat.

```
root@SHOFI2SIEUbuntu:~# sudo netplan apply
```

```
root@SHOFI2SIEUbuntu:~#
```

\$ ip a : mengecek hasil konfigurasi jaringan

```
root@SHOFI2SIEUbuntu:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b6:61:72 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feb6:6172/64 scope link
        valid_lft forever preferred_lft forever
root@SHOFI2SIEUbuntu:~#
```

\$ ping 8.8.8.8 : untuk **menguji apakah jaringan sudah terhubung ke internet**. [8.8.8.8 adalah DNS publik milik Google.] Koneksi internet berbasis IP

```
root@SHOFI2SIEUbuntu:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=11.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=12.9 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=16.0 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=11.9 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=25.4 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=17.7 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=255 time=14.8 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=255 time=15.1 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=255 time=11.1 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=255 time=16.7 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=255 time=13.1 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=255 time=19.5 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=255 time=11.3 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=255 time=16.4 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=255 time=14.4 ms
64 bytes from 8.8.8.8: icmp_seq=16 ttl=255 time=20.8 ms
64 bytes from 8.8.8.8: icmp_seq=17 ttl=255 time=12.0 ms
```

\$ ping -c 4 8.8.8.8 (menggunakan -c 4 agar berhenti stlh 4 baris jalan) :

-c 4 : Ping dibatasi 4 kali (lalu setelah 4x akan auto keluar), tidak terus menerus

```
root@SHOFI2SIEUbuntu:~# ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=11.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=11.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=11.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=47.8 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 11.201/20.451/47.826/15.804 ms
root@SHOFI2SIEUbuntu:~#
```

\$ ping -c 4 google.com : untuk menguji Koneksi internet + DNS

```

root@SHOFI2SIEUbuntu:~# ping -c 4 google.com
PING google.com (64.233.170.100) 56(84) bytes of data.
64 bytes from sg-in-f100.1e100.net (64.233.170.100): icmp_seq=1 ttl=255 time=16
.8 ms
64 bytes from sg-in-f100.1e100.net (64.233.170.100): icmp_seq=2 ttl=255 time=16
.5 ms
64 bytes from sg-in-f100.1e100.net (64.233.170.100): icmp_seq=3 ttl=255 time=10
.8 ms
64 bytes from sg-in-f100.1e100.net (64.233.170.100): icmp_seq=4 ttl=255 time=30
.5 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 10.816/18.637/30.522/7.258 ms
root@SHOFI2SIEUbuntu:~#

```

Percobaan 4 : VLAN tanpa switch CISCO

[VLAN adalah Virtual Local Area Network]

\$ ip a show enp0s3 : Untuk melihat kondisi interface jaringan

```

root@SHOFI2SIEUbuntu:~# ip a show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:b6:61:72 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feb6:6172/64 scope link
        valid_lft forever preferred_lft forever
root@SHOFI2SIEUbuntu:~#

```

\$ sudo apt update : Memperbarui daftar paket

Sebelum menginstall fitur tambahan VLAN, harus memperbarui daftar paket

```

root@SHOFI2SIEUbuntu:~# sudo apt update
Hit:1 http://id.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu focal-security InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
All packages are up to date.
root@SHOFI2SIEUbuntu:~#

```

\$ sudo apt install vlan : Menginstall dukungan VLAN di Linux

```

root@SHOFI2SIEUbuntu:~# sudo apt install vlan -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  vlan
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 11.3 kB of archives.
After this operation, 50.2 kB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu focal-updates/universe amd64 vlan all
2.0.4ubuntu1.20.04.1 [11.3 kB]
Fetched 11.3 kB in 0s (71.7 kB/s)
Selecting previously unselected package vlan.
(Reading database ... 180634 files and directories currently installed.)
Preparing to unpack .../vlan_2.0.4ubuntu1.20.04.1_all.deb ...
Unpacking vlan (2.0.4ubuntu1.20.04.1) ...
Setting up vlan (2.0.4ubuntu1.20.04.1) ...
Processing triggers for man-db (2.9.1-1) ...
root@SHOFI2SIEUbuntu:~#

```

\$ sudo modprobe 8021q : Mengaktifkan kemampuan VLAN di Linux agar komputer bisa membuat dan mengelola VLAN

```

root@SHOFI2SIEUbuntu:~# sudo modprobe 8021q
root@SHOFI2SIEUbuntu:~#

```


INTERFACE VLAN 10

\$ sudo ip link add link enp0s3 name enp0s3.10 type vlan id 10 (vlan 10 karna agar bagian itu terfokus bagian 10 aja) : Untuk **membuat jalur khusus** (VLAN 10) di atas jaringan utama enp0s3.

```
root@SHOFI2SIEUbuntu:~# sudo ip link add link enp0s3 name enp0s3.10 type vlan id 10
```

\$ sudo ip addr add 10.0.10.15/24 dev enp0s3.10 : Untuk **memberi alamat** ke VLAN 10 supaya bisa dikenali di jaringan.

```
root@SHOFI2SIEUbuntu:~# sudo ip addr add 10.0.10.15/24 dev enp0s3.10
root@SHOFI2SIEUbuntu:~#
```

\$ sudo ip link set enp0s3.10 up : Untuk **menghidupkan jalur VLAN 10** agar siap dipakai.

\$ ping 10.0.10.15 : Untuk mengetes apakah interface VLAN 10 sudah berfungsi.

```
root@SHOFI2SIEUbuntu:~# sudo ip link set enp0s3.10 up
root@SHOFI2SIEUbuntu:~# ping 10.0.10.15
PING 10.0.10.15 (10.0.10.15) 56(84) bytes of data.
 64 bytes from 10.0.10.15: icmp_seq=1 ttl=64 time=0.020 ms
 64 bytes from 10.0.10.15: icmp_seq=2 ttl=64 time=0.046 ms
 64 bytes from 10.0.10.15: icmp_seq=3 ttl=64 time=0.059 ms
 64 bytes from 10.0.10.15: icmp_seq=4 ttl=64 time=0.083 ms
 64 bytes from 10.0.10.15: icmp_seq=5 ttl=64 time=0.047 ms
```

COBA KALO VLAN 20

\$ sudo ip link add link enp0s3 name enp0s3.20 type vlan id 20 : Untuk membuat jalur baru lagi (VLAN 20) di atas enp0s3.

\$ sudo ip addr add 10.0.20.15/24 dev enp0s3.20 : Untuk memberi alamat ke VLAN 20.

\$ sudo ip link set enp0s3.20 up : Untuk **menghidupkan VLAN 20**.

\$ ping 10.0.20.15 : Untuk **ngecek apakah VLAN 20 sudah berfungsi**.

```
root@SHOFI2SIEUbuntu:~# sudo ip link add link enp0s3 name enp0s3.20 type vlan id 20
root@SHOFI2SIEUbuntu:~# sudo ip addr add 10.0.20.15/24 dev enp0s3.20
root@SHOFI2SIEUbuntu:~# sudo ip link set enp0s3.20 up
root@SHOFI2SIEUbuntu:~# ping -c 3 10.0.20.15
PING 10.0.20.15 (10.0.20.15) 56(84) bytes of data.
 64 bytes from 10.0.20.15: icmp_seq=1 ttl=64 time=0.018 ms
 64 bytes from 10.0.20.15: icmp_seq=2 ttl=64 time=0.139 ms
 64 bytes from 10.0.20.15: icmp_seq=3 ttl=64 time=0.053 ms

--- 10.0.20.15 ping statistics ---
 3 packets transmitted, 3 received, 0% packet loss, time 2040ms
 rtt min/avg/max/mdev = 0.018/0.070/0.139/0.050 ms
root@SHOFI2SIEUbuntu:~#
```

\$ ip a show enp0s3 : Untuk **melihat informasi lengkap** tentang interface jaringan enp0s3

```
root@SHOFI2SIEUbuntu:~# ip a show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b6:61:72 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feb6:6172/64 scope link
        valid_lft forever preferred_lft forever
root@SHOFI2SIEUbuntu:~#
```

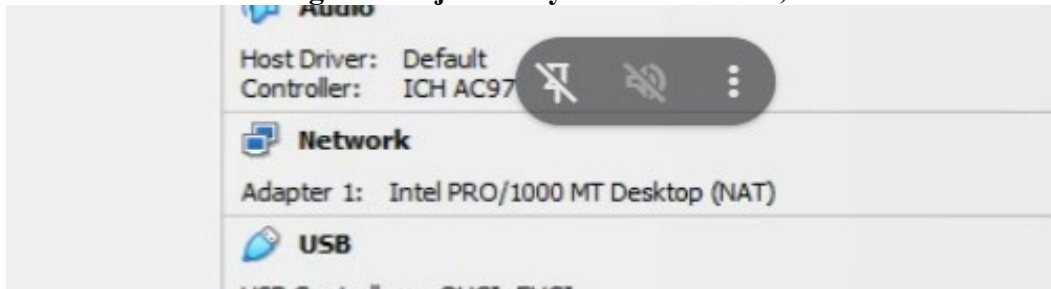
\$ ping 10.0.20.50 -I enp0s3.10 : **mengirim ping dari VLAN 10** menuju alamat yang ada di **VLAN 20**

```
root@SHOFI2SIEUbuntu:~# ping 10.0.20.15 -I enp0s3.10
PING 10.0.20.15 (10.0.20.15) from 10.0.10.15 enp0s3.10: 56(84) bytes of data.
From 10.0.10.15 icmp_seq=1 Destination Host Unreachable
From 10.0.10.15 icmp_seq=2 Destination Host Unreachable
From 10.0.10.15 icmp_seq=3 Destination Host Unreachable
```

Karena **VLAN 10 dan VLAN 20** adalah **jaringan yang terpisah**, jadi mereka **tidak bisa saling mengakses** tanpa router/inter-VLAN routing.

Percobaan 5 : Routing Antar VLAN (InterVLAN Routing)

*NOTE : tidak bisa digunakan jika hanya 1 VM Ubuntu, harus 2 VM Ubuntu



Saran bg mickel : Sebelum jalankan ubuntu, pada bagian *Network* ubah NAT jadi Host-Only

Ini gunakan NAT (dilab)

\$ sudo sysctl -w net.ipv4.ip_forward=1 : Untuk mengaktifkan fitur routing di Linux supaya bisa meneruskan paket antar jaringan/VLAN.

```
root@SHOFI2SIEubuntu:~# sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
root@SHOFI2SIEubuntu:~#
```

\$ ping 8.8.8.8 -I enp0s3.10 : Untuk mengetes apakah VLAN 10 bisa keluar ke internet lewat interface enp0s3.10.

Hasilnya : *Destination Host Unreachable* : artinya VLAN 10 belum bisa keluar internet karena tidak ada gateway atau NAT.

```
root@SHOFI2SIEubuntu:~# ping 8.8.8.8 -I enp0s3.10
PING 8.8.8.8 (8.8.8.8) from 10.0.10.15 enp0s3.10: 56(84) bytes of data.
From 10.0.10.15 icmp_seq=1 Destination Host Unreachable
From 10.0.10.15 icmp_seq=2 Destination Host Unreachable
From 10.0.10.15 icmp_seq=3 Destination Host Unreachable
From 10.0.10.15 icmp_seq=4 Destination Host Unreachable
From 10.0.10.15 icmp_seq=5 Destination Host Unreachable
From 10.0.10.15 icmp_seq=6 Destination Host Unreachable
From 10.0.10.15 icmp_seq=7 Destination Host Unreachable
From 10.0.10.15 icmp_seq=8 Destination Host Unreachable
From 10.0.10.15 icmp_seq=9 Destination Host Unreachable
^C
--- 8.8.8.8 ping statistics ---
10 packets transmitted, 0 received, +9 errors, 100% packet loss, time 9195ms
pipe 4
root@SHOFI2SIEubuntu:~#
```

\$ ping 8.8.8.8 -I enp0s3.20 : Untuk mengetes apakah VLAN 20 bisa akses internet lewat interface enp0s3.20.

Hasilnya : *Destination Host Unreachable* : Sama seperti VLAN 10, VLAN 20 juga tidak bisa keluar internet.

```
root@SHOFI2SIEubuntu:~# ping 8.8.8.8 -I enp0s3.20
PING 8.8.8.8 (8.8.8.8) from 10.0.20.15 enp0s3.20: 56(84) bytes of data.
From 10.0.20.15 icmp_seq=1 Destination Host Unreachable
From 10.0.20.15 icmp_seq=2 Destination Host Unreachable
From 10.0.20.15 icmp_seq=3 Destination Host Unreachable
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 0 received, +3 errors, 100% packet loss, time 4082ms
pipe 4
```


Gunakan “Host-only Adapt”(di laptop rumah)



Network

Adapter 1: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter')

- VLAN tetap **tidak bisa** ping internet, *karena* Ini adalah **VirtualBox Host-Only Adapter**, dan karakteristiknya adalah Tidak bisa terhubung ke internet, Tidak punya gateway, Tidak bisa route ke luar host dan hanya bisa komunikasi dengan host (komputer fisik)

```
root@SHOFI2SIEUbuntu:~# ping 8.8.8.8 -I enp0s3.10
PING 8.8.8.8 (8.8.8.8) from 192.168.10.101 enp0s3.10: 56(84) bytes of data.
From 192.168.10.101 icmp_seq=1 Destination Host Unreachable
From 192.168.10.101 icmp_seq=2 Destination Host Unreachable
From 192.168.10.101 icmp_seq=3 Destination Host Unreachable
From 192.168.10.101 icmp_seq=4 Destination Host Unreachable
```

Percobaan 5 ganti jadi Loopback VLAN

```
3: enp0s3.10@enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue s
tate UP group default qlen 1000
    link/ether 08:00:27:94:59:fc brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.101/24 scope global enp0s3.10
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe94:59fc/64 scope link
        valid_lft forever preferred_lft forever
4: enp0s3.20@enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue s
tate UP group default qlen 1000
    link/ether 08:00:27:94:59:fc brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.101/24 scope global enp0s3.20
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe94:59fc/64 scope link
        valid_lft forever preferred_lft forever
```

\$ **ip a** dulu, utk
lihat inet yg
dilaptop masing”

\$Ping -I enp0s3.10 192.168.10.101 (inet di ip a laptop)

```
root@SHOFI2SIEUbuntu:~# ping -I enp0s3.10 192.168.10.101
PING 192.168.10.101 (192.168.10.101) from 192.168.10.101 enp0s3.10: 56(84) byte
s of data.
64 bytes from 192.168.10.101: icmp_seq=1 ttl=64 time=0.926 ms
64 bytes from 192.168.10.101: icmp_seq=2 ttl=64 time=1.72 ms
64 bytes from 192.168.10.101: icmp_seq=3 ttl=64 time=0.102 ms
^C
--- 192.168.10.101 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2031ms
rtt min/avg/max/mdev = 0.102/0.915/1.718/0.659 ms
```

\$Ping -I enp0s3.20 192.168.20.101 (inet di ip a laptop)

```
root@SHOFI2SIEUbuntu:~# ping -I enp0s3.20 192.168.20.101
PING 192.168.20.101 (192.168.20.101) from 192.168.20.101 enp0s3.20: 56(84) byte
s of data.
64 bytes from 192.168.20.101: icmp_seq=1 ttl=64 time=1.04 ms
64 bytes from 192.168.20.101: icmp_seq=2 ttl=64 time=0.046 ms
64 bytes from 192.168.20.101: icmp_seq=3 ttl=64 time=0.102 ms
64 bytes from 192.168.20.101: icmp_seq=4 ttl=64 time=0.048 ms
```

\$Ping -c 3 192.168.20.101 -I 192.168.10.101

```
root@SHOFI2SIEUbuntu:~# ping -c 3 192.168.20.101 -I 192.168.10.101
PING 192.168.20.101 (192.168.20.101) from 192.168.10.101 : 56(84) bytes of data
.
64 bytes from 192.168.20.101: icmp_seq=1 ttl=64 time=0.760 ms
64 bytes from 192.168.20.101: icmp_seq=2 ttl=64 time=0.032 ms
64 bytes from 192.168.20.101: icmp_seq=3 ttl=64 time=0.081 ms

--- 192.168.20.101 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2030ms
rtt min/avg/max/mdev = 0.032/0.291/0.760/0.332 ms
```

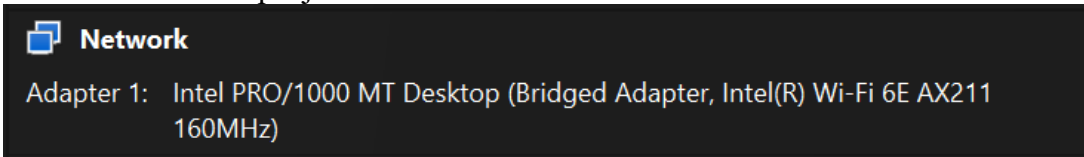
LATIHAN

1. Lakukan analisis mengapa ketika melakukan ping antara VLAN10 dan VLAN20 terjadi keterangan:

```
From 192.168.10.101 icmp_seq=1 Destination Host Unreachable
From 192.168.10.101 icmp_seq=2 Destination Host Unreachable
From 192.168.10.101 icmp_seq=3 Destination Host Unreachable
From 192.168.10.101 icmp_seq=4 Destination Host Unreachable
```

- Ping antar VLAN gagal karena **VLAN10 dan VLAN20 berada di jaringan yang berbeda dan terisolasi**, sehingga **tidak ada jalur (routing) yang menghubungkan keduanya**.
- Secara default, Linux **tidak melakukan routing antar subnet**. Akibatnya, saat VLAN10 mencoba ping VLAN20, paket tidak tahu harus lewat mana.

2. Silahkan cari solusi agar bisa melakukan ping antar VLAN, nanti hasilnya silahkan di screenshot dan beri penjelasan.



- Pakai Host-only = tidak bisa
- Pakai NAT tidak bisa
- Pakai bridged adapt = tidak bisa

```
root@SHOFI2SIEUbuntu:~# ping 192.168.20.101 -I enp0s3.10
PING 192.168.20.101 (192.168.20.101) from 192.168.10.101 enp0s3.10: 56(84) byte
s of data.
From 192.168.10.101 icmp_seq=1 Destination Host Unreachable
From 192.168.10.101 icmp_seq=2 Destination Host Unreachable
From 192.168.10.101 icmp_seq=3 Destination Host Unreachable
```

Berdasarkan searching, hasilnya

Wi-Fi TIDAK mendukung VLAN (802.1Q)

VirtualBox *tidak bisa* melewati frame VLAN melalui WiFi, sehingga:

- VLAN 10 dan VLAN 20 memang bisa muncul di Ubuntu
- TAPI paket VLAN tidak pernah keluar NIC
- Semua ping antar VLAN akan tetap → Destination Host Unreachable

Ini bukan salah konfigurasi, tapi keterbatasan hardware WiFi.

Solusi yang disarankan :

- **Gunakan Ethernet LAN : Colok kabel LAN ke laptop → VirtualBox → Network → Bridged Adapter → pilih Ethernet.**
- **Gunakan USB-to-LAN Adapter**
- **Kerjakan di komputer LAB : Komputer lab pakai LAN → VLAN pasti berjalan.**

- Harus **menggunakan 2 VM Ubuntu**